

GuardianX:

**AUTONOMOUS AI-DRIVEN HUMAN DETECTION
AND ROOM-PRIORITY RESCUE TRIAGE IN
HIGH-RISE BUILDING FIRES**

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1. Abstract:

High-rise and complex building fires claim numerous lives each year, and initial responders often lack immediate, organized intelligence regarding the location of victims and active hazards. In this project, I present a software minimum viable product (MVP) that I designed to optimize search-and-rescue triage and situational awareness in structural firefighting environments. By utilizing a YOLO-based object detection model, the system analyzes images from interior rooms to accurately identify humans, fire, and smoke. Rather than outputting raw, uncontextualized confidence scores, the software processes this visual evidence into a tactical triage dashboard that ranks rooms based on logical states and victim volume. These classifications directly align with emergency dispatch protocols, prioritizing rooms from critical multi-victim scenarios (life threat combined with active fire) down to lower-tier suppression zones (active fire with no visible victims). Ultimately, this system does not make autonomous rescue decisions; instead, it empowers incident command with organized, actionable data, reducing emergency response latency and allowing firefighters to rapidly direct Search & Rescue (SAR) and Suppression teams where they are needed most.

Keywords: Computer Science and Software Engineering; Artificial Intelligence; Computer Vision; Human Detection in Smoke; High-Rise Building Fire Rescue; Autonomous UAV

2. Introduction:

2.1 (Problem Statement)

High-rise building fires in densely populated urban areas claim numerous lives each year, and the absence of AI-driven autonomous rescue systems forces firefighters to operate under extreme conditions of smoke, heat, and structural collapse in very dynamic fire scenes. Traditional visual search and reconnaissance methods frequently fail due to heavy smoke opacity and dense debris, hence the incident commander and his team need to spend extra time and resources in order to identify and rescue trapped individuals, especially in high-rise buildings.

2.2 (General Context)

The integration of artificial intelligence (AI) and computer vision is transforming fire rescue operations, particularly in hazardous high-rise building scenarios where immediate situational awareness is heavily restricted. During structural fires, low visibility due to smoke, heat, and structural instability creates massive challenges for incident commanders trying to locate trapped individuals. Recent research demonstrates that deep learning models—specifically YOLO variants and Convolutional Neural Networks (CNNs)—can rapidly identify and localize humans in these dangerous environments by processing interior imagery. Ongoing research highlights how these models operate effectively under fire conditions. For instance, the study “*Detection of Human Life in Fire Scenarios Using YOLOv8*” demonstrates the model's ability to utilize OpenCV frameworks to successfully isolate human positions within bounding boxes, even in complex scenes. While this paper provides a robust computational subsystem, its findings yield raw detection data (such as confidence scores) rather than actionable tactical intelligence. In a life-or-death scenario, raw numbers are insufficient for emergency dispatchers who must immediately decide where to send specific teams. I bridge this gap by proposing a software Minimum Viable Product (MVP) focused on realistic decision support rather than autonomous robotics. By analyzing images from various environments to detect humans, fire, and smoke, the system synthesizes visual evidence into an action-oriented triage dashboard. Instead of making the final rescue decision, the software organizes operational intelligence by ranking rooms based on rescue urgency and logical states—prioritizing multi-victim critical zones requiring immediate Search & Rescue (SAR) overactive fire zones requiring Suppression teams. This approach synthesizes the latest advancements in AI detection models with practical, human-in-the-loop disaster response protocols.

2.3 (Aim and objectives)

My aim of this project is to develop GuardianX, a tactical decision-support dashboard that uses a YOLOv8-OBB (Oriented Bounding Box) model to detect humans, fire, and smoke inside buildings during structural fires. The system converts raw visual detection data into an automated, room priority rescue-triage dashboard, giving incident commanders immediate, actionable intelligence to improve search-and-rescue (SAR) response times.

I can achieve my aim when:

- The OBB model detects humans, fire, and smoke with high precision under low-visibility and heavy smoke conditions.
- The system successfully classifies rooms into structured triage priority tiers based on victim count and active hazards.
- The dashboard displays real-time annotated visual feeds alongside clear dispatch recommendations.
- The application reduces emergency response latency by converting raw confidence scores into logical room rankings.

Objectives

- Make a YOLOv8-OBb model for robust object detection of humans, fire, and smoke in high-temperature, low-visibility environments.
- Formulate a rule-based triage algorithm that translates detection outputs into room priority classifications.
- Develop a high-performance web dashboard (using Streamlit) for real-time visualization and incident command monitoring.
- Program an automated dispatch recommender system to generate actionable alerts for Search & Rescue (SAR) and Fire Suppression teams.
- Validate the system's accuracy and latency using benchmark test datasets and simulated building fire feeds.

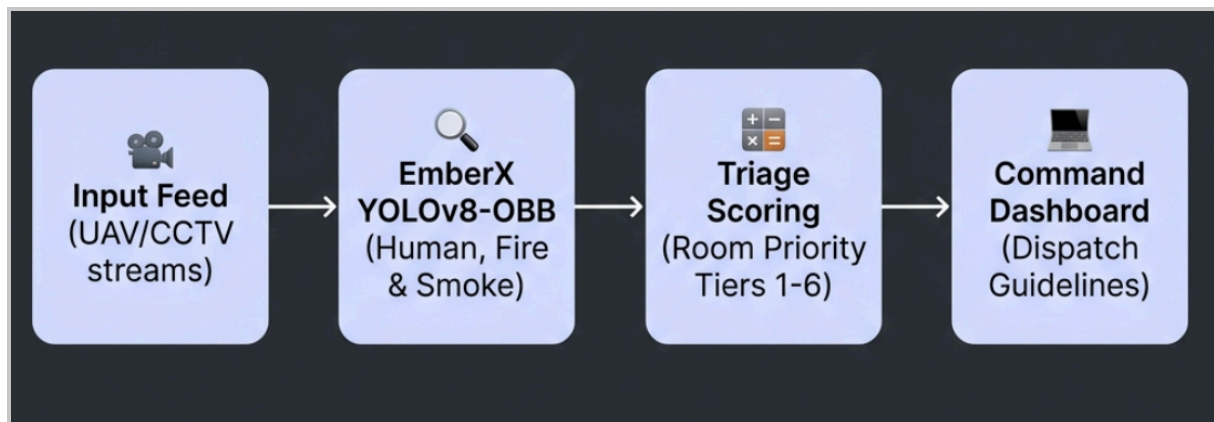
3. Methods:

A systematic literature search targeted computer vision models for detecting humans, fire, and smoke in low visibility environments, as well as AI driven search-and-rescue (SAR) frameworks and UAV operations for disaster response, and sought existing dataset benchmarks, edge cases, and technical limitations. From an initial pool of 7,212 publications, strict relevance filtering and deduplication yielded a final corpus of 50 peer reviewed papers.

3.1 System Input Specifications:

- Source: Video/image feed from unmanned aerial vehicles (UAVs), indoor CCTV cameras.
- Resolution: Standard resolution with near appropriate lighting.
- Colour Space: 3-channel RGB.

GuardianX pipeline overview:



3.2 Engineering Methodology:

- Object Detection: The frame is fed into my trained YOLOv8-OBB model, which predicts oriented bounding boxes for three classes: human, fire, and smoke.
- Prioritization Score Calculation: Using the detection counts, my triage algorithm computes a room-priority score.

$$\text{Priority Score} = (0.55 * H \text{ det}) + (0.25 * F \text{ det}) + (0.15 * S \text{ det}) + (0.05 * H \text{ count})$$

Where:

- $H \text{ det} = (H \text{ count} \geq 1) ? 1 : 0$
 - $F \text{ det} = (F \text{ count} \geq 1) ? 1 : 0$
 - $S \text{ det} = (S \text{ count} \geq 1) ? 1 : 0$
- User Interface Render: The dashboard updates the room's status, displays the video feed with custom bounding box overlays, and prints dispatch instructions.

3.3 The process of Data collection to Model training

For this project I began training the model on Google Colab using a RoboFlow dataset called “fire human smoke detector”. Although the dataset originates from RoboFlow, I ran the training on Colab with in-built GPU support. After about 20–30 minutes each epoch failed because the available computational resources were insufficient. This was a major drawback, as I needed more epochs and more time.

I then switched to the Kaggle notebooks, which offered better computational power. However, because the data came from RoboFlow, I struggled to import the dataset into the notebook. While the notebooks solved the resource issue, several required libraries could not be installed on Kaggle, presenting another obstacle.

Finally, I explored RoboFlow's cloud-based training. Since the training runs on RoboFlow's servers, it is not limited by my local computational resources. Fortunately, I had enough credits to train the model for a sufficient number of epochs, achieving the desired accuracy. Another important aspect of model training is image augmentation. While training with RoboFlow, I learned how crucial augmentation is for dynamic fire-scene environments, ensuring the model can learn from a dataset of dynamic images.

RoboFlow's augmentation increased the dataset from about 9.7 k images to 17 k images, making it more robust. This improved training, although the cloud-based process was slow, training took roughly nine hours. Nevertheless, the model succeeded.

4. Wider Purpose

4.1. Enhancing Life Safety in High-Rise Urban Fires

High-rise structure fires present severe egress and reconnaissance challenges, where delayed search-and-rescue (SAR) operations directly increase victim mortality from toxic smoke inhalation and thermal exposure. By utilizing real-time object detection, rescue teams can bypass manual sweep procedures and immediately target high-risk areas (e.g., active fires with trapped victims), accelerating extraction times and saving lives in time-critical windows. This saves time and resources for the incident commander and his team

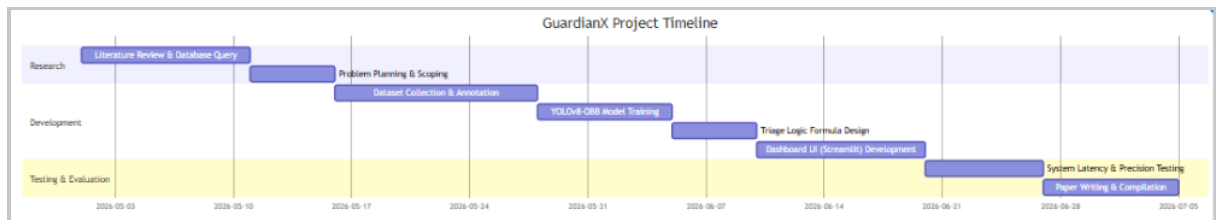
4.2. First Responder Protection and Resource Optimization

Traditional fire rescue teams may take the quickest route at first but at different places and times the route is not always known to the navigation system, in terms of sudden road blocks or huge traffic conditions. GuardianX acts as a drone as first responder (DFR). This not only informs the fire department the fastest route but also results in an instant deployment before the incident commander and his team could even leave the facility. With this DFR technology, fire fighters can prepare for the dynamic scene they are about to witness beforehand, instead of going into the scene almost blind.

5. Project Planning, Scope, and Timeline:

In the initial planning phase of this project, I defined the scope to address the information bottleneck at the incident command level. I structured the planning phase into three major steps:

- **Constraint Analysis:** Assessing the limitations of standard RGB cameras in smoke-filled environments.
- **Algorithm Design:** Designing a logic framework that translates raw object detections into a priority queue.
- **Dashboard Prototyping:** Developing a real-time Streamlit dashboard designed to present recommendations to incident commanders within seconds of a feed being established.



6. Range of Approaches:

1. Attempt to train model in Google Colab with YOLO_v8 with OBB

Trained on integrated GPU system but the computational power was insufficient for completing the desired number of epochs and it failed half way through the training, hence I could not get the desired model expectation

2. Attempt to train model in Kaggle notebook

The training was unsuccessful on Kaggle as well because the Roboflow dataset failed to upload for model training. This meant the code would strike errors and the model can't fully reach my desired output

3. A successful training on Roboflow cloud based Platform

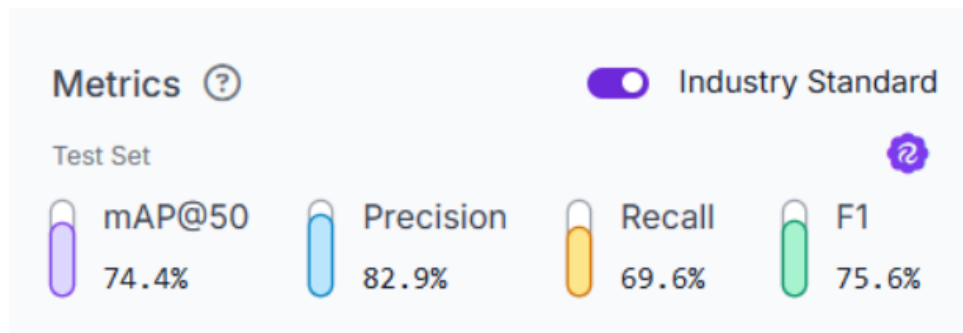
The training was completed on the Roboflow platform itself as this required only credits and not literal "occupational units", this was a result of cloud based training. I got to modify the dataset by adding images which were augmented to match a real scene. This model was trained under YOLO_v8 OBB, ending with desired results.

7. Results:

7.1 Core Technologies for Human Detection & Rescue

Recent frameworks combine Building Information Modelling (BIM), real-time image analysis via deep learning (e.g., transfer AlexNet), and neural network-based route planning to generate safe evacuation paths with high localization accuracy (<1.5 m error). Thermal imaging combined with CNNs (Ultralytics and YOLO) enables >80% efficiency in victim identification even under low-light, debris-filled conditions and dense smoke. Fusion of RGB cameras with the room rank protocol enables search and rescue (SAR) missions to be conducted more efficiently, and fire suppression resources are allocated faster.

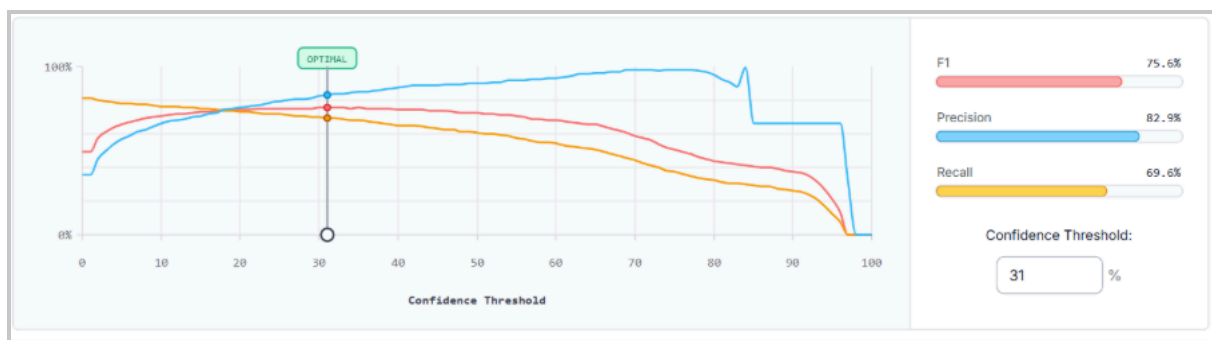
Reference - [Roboflow](#)



7.2 Model Performance: *EmberX (YOLOv8)*

The custom-trained EmberX object detection model, built on the YOLOv8 architecture, was evaluated on a fire-smoke ground test set. The model achieved a Precision of 82.9% (minimizing false alarms for rescue teams) and a Recall of 69.6% (reliably identifying victims and hazards despite smoke occlusion), yielding a balanced F1-score of 75.6%. Additionally, the model demonstrated robust target localization with a mAP@50 of 74.4% under complex, low-visibility conditions.

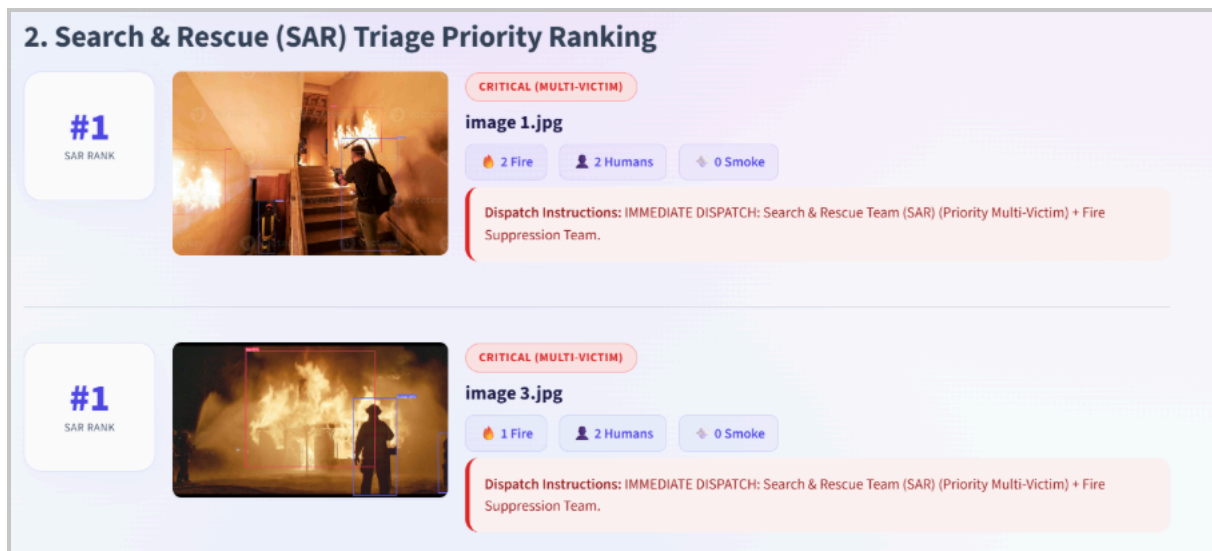
Reference - [Roboflow](#)



6.3 Triage Dashboard Interface

The tactical dashboard translates raw detections into an action-oriented interface for incident commanders. For each monitored feed, it displays:

- **Urgency Ranks & Severity Badges** (e.g., #1 SAR RANK, CRITICAL (MULTI-VICTIM)) to highlight immediate needs.
- **Annotated Visual Feeds** showing bounding box overlays of active fire and located victims.
- **Detection Metric Pills** providing quick counts of detected fire, humans, and smoke.
- **Actionable Dispatch Prompts** delivering clear, automated instructions directly to rescue and suppression teams.



7.3 Robotics Platforms & Navigation

In this system design, self-contained robotic platforms such as indoor UAVs serve as data-acquisition tools by capturing real-time RGB video. Instead of resource-intensive SLAM mapping, the video is processed with the YOLOv8-OBb algorithm, which detects humans, fire, and smoke. These detections feed a triage algorithm that prioritizes rooms.

8. Discussions:

The reviewed literature confirms that AI-powered autonomous systems can significantly enhance human identification and rescue operations during high-rise fires by leveraging computer vision (especially thermal imaging), deep learning classification models (YOLO variants), sensor fusion (RGB + LiDAR/Thermal), and real-time navigation algorithms. These technologies reduce responder risk by enabling robots/UAVs to operate independently in hazardous zones. However, deployment at scale faces hurdles: sensor degradation under extreme conditions remains a concern; generalization across building types/environments is not fully solved; annotated datasets specific to smoke-obscured or cluttered scenes are still limited; edge computing constraints challenge real-time inference; ethical considerations around autonomy persist. Despite these challenges, the field is rapidly advancing—especially with open-source datasets now available for model training/testing

9. Conclusions

AI-driven autonomous solutions that combine computer vision and deep learning with the room-ranking protocol are effective at identifying and rescuing people during high-rise fires while minimizing risk to fire responders. Recent advances include robust victim identification in smoke or low-visibility conditions and a criteria-based environmental ranking for the incident commander. The robotic/UAV system handles dangerous tasks in hazardous zones; humans manage overall strategy, resource allocation, complex rescues, and final interventions. This ensures the AI-driven system coordinates with the fire department's team.

10. AI Acknowledgements:

In this paper, I have used Consensus AI to conduct a literature review on the topic. Google Antigravity was employed to develop the Incident Commander dashboard. Google Gemini assisted in coding the YOLO-based model. After using these tools, Kishan Kannan has reviewed and edited the content as needed and assumes full responsibility for the publication.

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